Supplementary Information

Bench-Scale Liquid-Phase Microaeration and Iron Dosing Comparison for In Situ Biogas Desulfurization and Antibiotic Resistance Management in Mesophilic Anaerobic Digestion

Seyed Mohammad Mirsoleimani Azizi[1], Basem S Zakaria[1], Nervana Haffiez[1], Parisa Niknejad[1], Rasha Maal-Bared[2], Saif Molla[2], Bipro Ranjan Dhar[1,*]

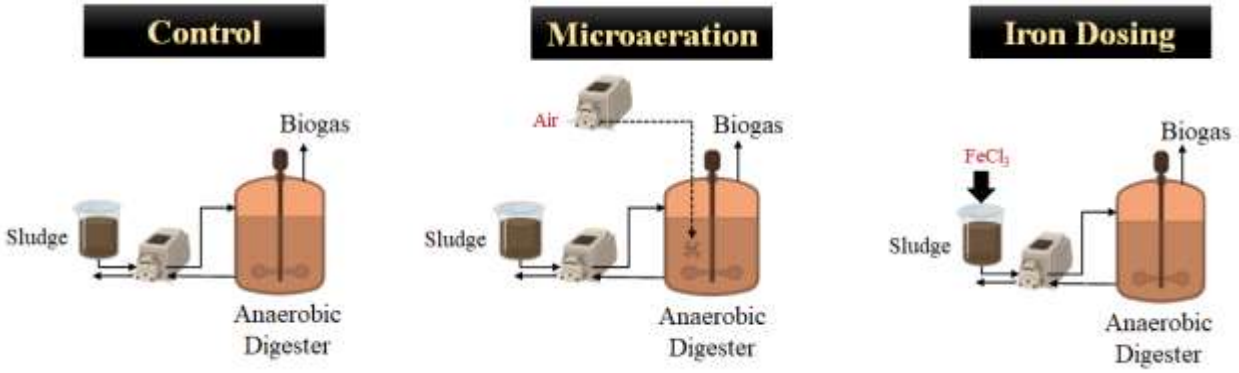
[1]Civil and Environmental Engineering, University of Alberta, 116 Street NW, Edmonton, AB T6G 1H9, Canada

[2] EPCOR Water Services, 2000–10423 101 Street NW, Edmonton, Alberta T5H 0E8, Canada

- **Number of pages:** 6
- **Number of figures:** 4
- **Number of tables:** 1

[*]Corresponding author. Bipro Ranjan Dhar (bipro@ualberta.ca)

(a)



(b)

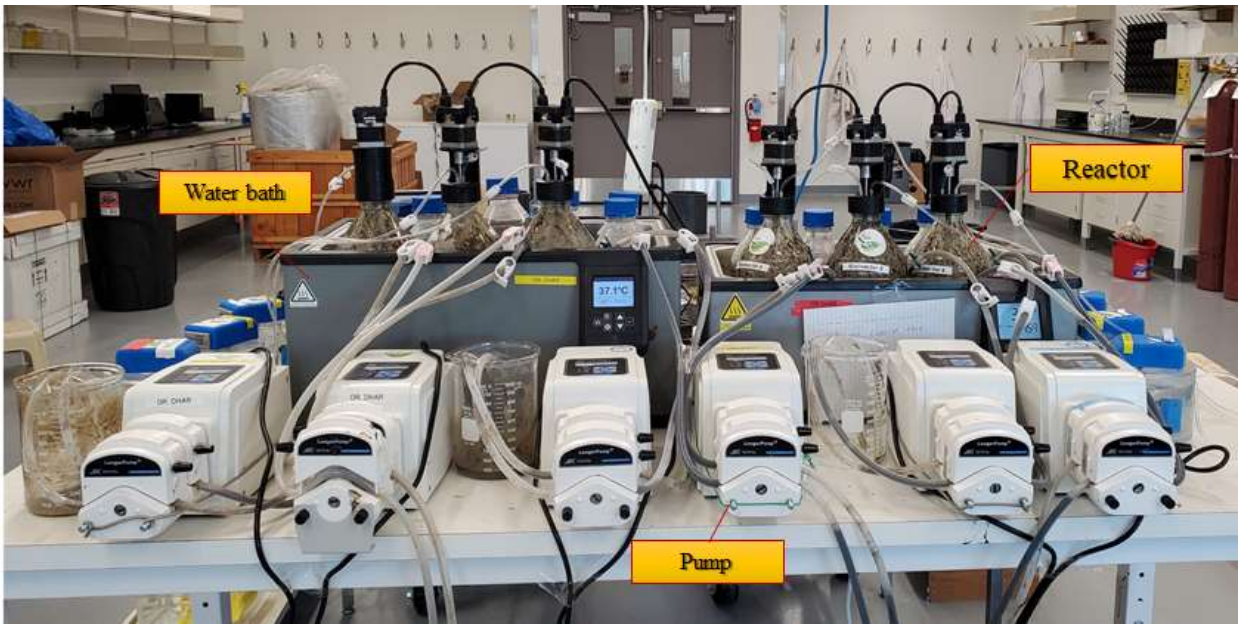
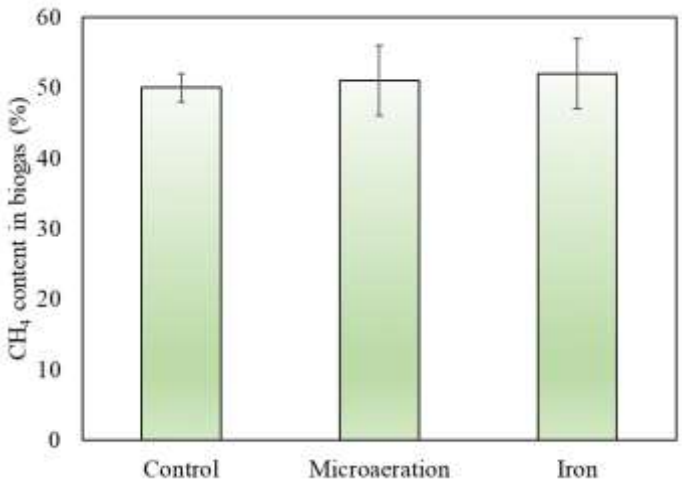


Fig. S1. (a) Schematic representation, and (b) photograph of experimental set-up.

(a)



(b)

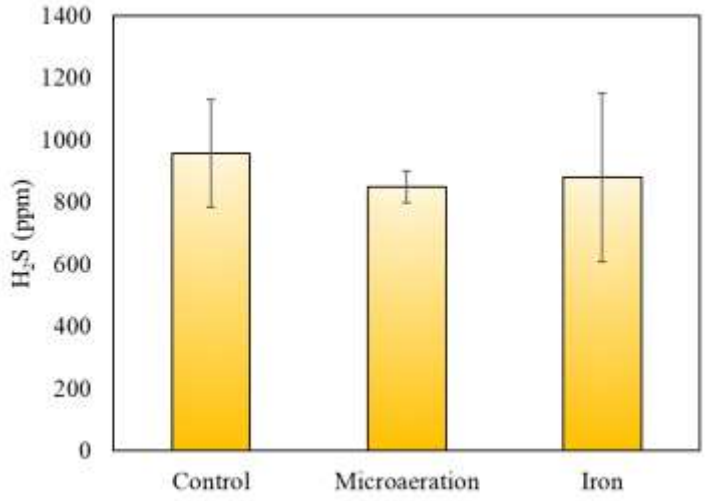
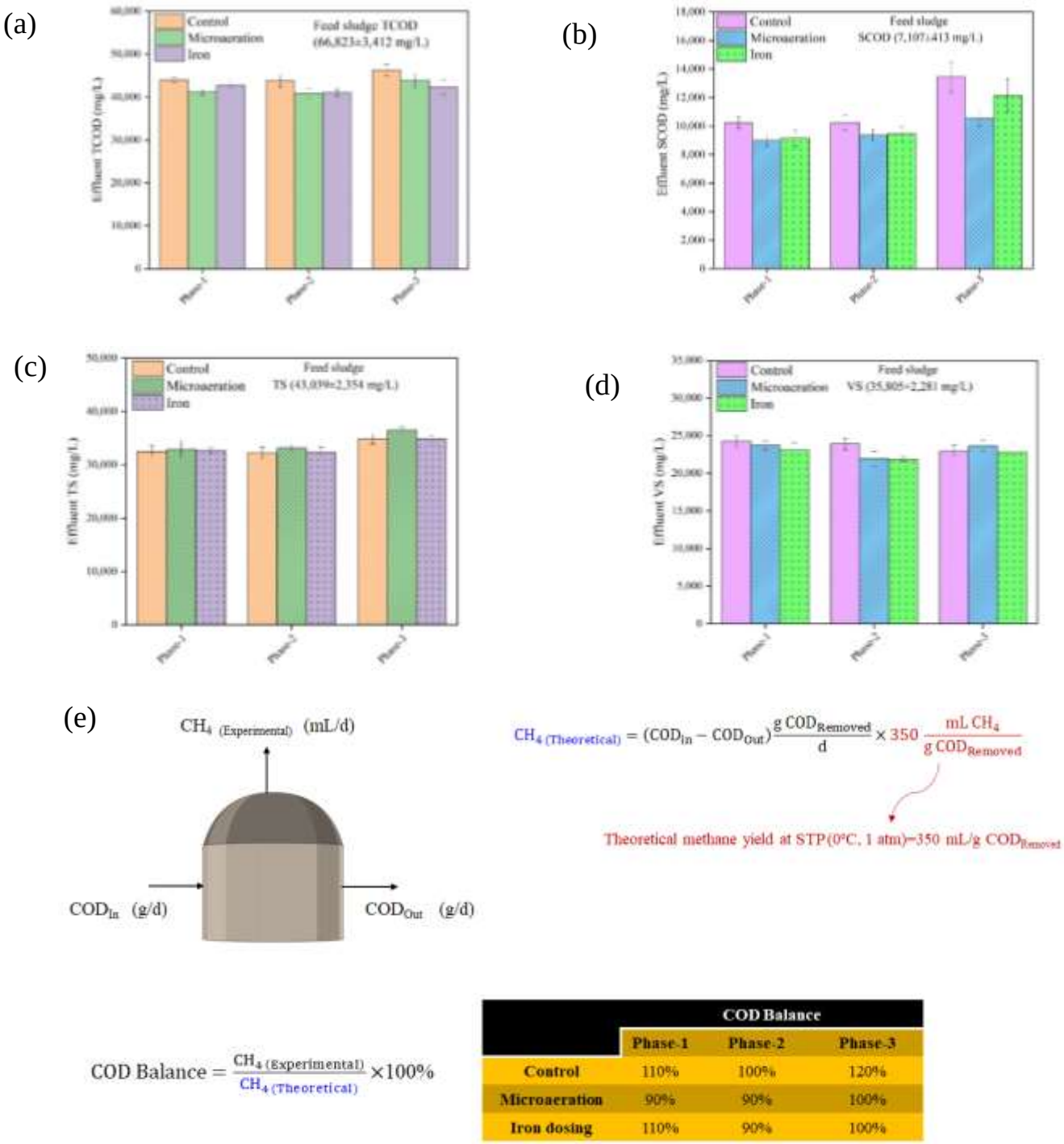


Fig. S2. (a) Methane, and (b) H_2S content in biogas measured during the pre-experimental phase.



	COD Balance		
	Phase-1	Phase-2	Phase-3
Control	110%	100%	120%
Microaeration	90%	90%	100%
Iron dosing	110%	90%	100%

Fig. S3. (a) Effluent TCOD, (b) effluent SCOD, (c) effluent TS, (d) effluent VS, and (e) COD balance under different experimental phases.

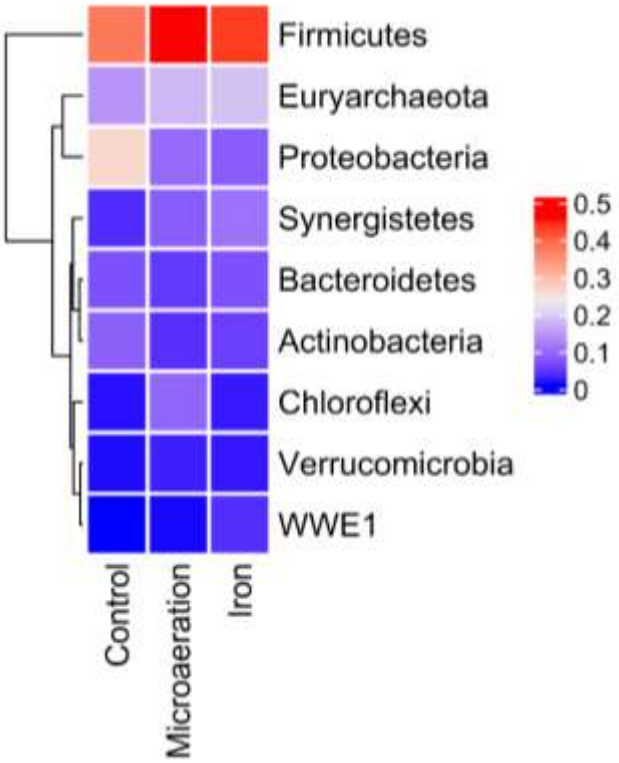


Fig. S4. Bacterial relative abundance at the phylum level in phase-3.

Table S1. Alpha diversity metrics of microbial community in different reactors in phase-3.

	Chao1 Species Richness	**Pielous's evenness Index**	**Observed Number of Microbial OTUs**	**Shannon-Wiener Diversity Index**	**Simpson's Reciprocal Index**
Control	174	0.92	181	6.8	0.91
Microaeration	150	0.83	160	6.2	0.87
Iron Dosing	128	0.84	132	6.3	0.85